

YI HUANG

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Research Interests

- Hardware-software co-design for energy-efficient AI
- Neuromorphic computing based on emerging memory devices
- Intelligent sensing and communication systems for edge applications

Professional Experience

University of Tennessee, Knoxville

August 2026 – present

Assistant Professor

University of Massachusetts Amherst

September 2025 – July 2026

Postdoctoral Research Associate

Education

University of Massachusetts Amherst

September 2025

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Qiangfei Xia

Huazhong University of Science and Technology

June 2019 / June 2016

M.S. in Control Science and Engineering / B.E. in System Engineering

Advisor: Prof. Zhigang Zeng

Honors and Awards

Ting-wei Tang Dissertation Prize of University of Massachusetts Amherst (2026)

ECE David H. Navon Scholarship of University of Massachusetts Amherst (2022)

ECE Graduate Fellowship of University of Massachusetts Amherst (2020)

Publications

Journal Articles (* indicate equal contribution)

1. Analog neural network–based wireless receiver
C. He*, **Y. Huang***, A. Korkmaz, A. Jaber Rad, Q. Xia, and L. Katehi
Science Advances, **12**, eaeg7153 (2026)
2. Single-step analog in-memory matrix computation in three-dimensional circuits
A. Jaber Rad, K. Kumar, F. Kiani, **Y. Huang**, A. Abdel-Maksoud, Z. Xiong, Y. Ling, W. Zhao, and Q. Xia
Advanced Intelligent Systems, e70500 (2026)
3. Hyperdimensional in-memory computing with analogue memristive crossbar arrays
Y. Huang*, A. J. Rad*, D. Belkin, N. Ge, J. Joshua Yang, M. Hu, and Q. Xia
Nature Communications, **17**, 9162 (2026)
4. Electrochromic hyperspectral embedding for ultracompact intelligent vision

- R. Li, C. He, **Y. Huang**, E. Zhai, V. K. Sangwan, J. Zou, K. St. Luce, J. Cui, R. Kim, R. Liu, Z. Wang, X. Ling, H. Xie, X. Zhang, M. C. Hersam, Q. Xia, L. Katehi, and Y. C. Lin
Nature Sensors, **1**, 443–456 (2026)
5. Memristive cellular neural networks for fast in-pixel computing
V. Ravichandran, **Y. Huang**, B. Flannery, T. Molom-Ochir, T. Maurer, S. Asapu, A. Abdel-Maksoud, N. Heermance, R. Yoo, J. Tackie, W. Zhao, Y. Ling, A. Guo, J.J. Yang, Q. Xia
Nature Electronics **9**, 189–199 (2026).
 6. Event-based neuromorphic sensing system with flexible haptic sensors and a memristive system on a chip
W. Zhao, **Y. Huang**, A. Tewari, A. J. Rad, A. Zhang, N. Ge, J.J. Yang, M. Hu, S. Majumdar, Q. Xia
Nature Sensors **1**, 163–171 (2026).
 7. Radiofrequency signal processing with a memristive system-on-a-chip
Y. Huang*, C. He*, Y. Ling, N. Ge, J.J. Yang, M. Hu, L. Katehi, Q. Xia
Nature Electronics **8**, 587–596 (2025). **(Cover article, July 2025)**
 8. A parallel read-write circuit with fast amplitude-adaptive matching scheme to memristor crossbar array
N. Jiang, M. Jiang, H. Huang, L. Yang, **Y. Huang**, Z. Zeng
IEEE Transactions on Circuits and Systems I: Regular Papers **72**, 5898–5911 (2025).
 9. Memristor-based hardware accelerators for artificial intelligence
Y. Huang, T. Ando, A. Sebastian, M.F. Chang, J.J. Yang, Q. Xia
Nature Reviews Electrical Engineering **1**, 286–299 (2024).
 10. From memristive devices to neuromorphic systems
Y. Huang, F. Kiani, F. Ye, Q. Xia
Applied Physics Letters **122**, 110501 (2023). **(Editor's Pick)**
 11. Towards energy-efficient computing hardware based on memristive nanodevices
Y. Huang, V. Ravichandran, W. Zhao, Q. Xia
IEEE Nanotechnology Magazine **17**, 30–38 (2023).
 12. MERRC: A memristor-enabled reconfigurable low-power reservoir computing architecture at the edge
F. Nowshin, **Y. Huang**, M.R. Sarkar, Q. Xia, Y. Yi
IEEE Transactions on Circuits and Systems I: Regular Papers **71**, 174–186 (2023).
 13. Diffusive memristors with uniform and tunable relaxation time for spike generation in event-based pattern recognition
F. Ye, F. Kiani, **Y. Huang**, Q. Xia
Advanced Materials **35**, 2204778, (2022).
 14. Memristor-based HTM spatial pooler with on-device learning for pattern recognition
X. Liu, **Y. Huang**, Z. Zeng, D.C. Wunsch
IEEE Transactions on Systems, Man, and Cybernetics: Systems **52**, 1901–1915 (2020).
 15. An associative-memory-based reconfigurable memristive neuromorphic system with synchronous weight training
L. Yang, Z. Zeng, **Y. Huang**
IEEE Transactions on Cognitive and Developmental Systems **12**, 529–540 (2020).

16. Memristor-based circuit design for neuron with homeostatic plasticity
X. Shi, Z. Zeng, L. Yang, **Y. Huang**
IEEE Transactions on Emerging Topics in Computational Intelligence **2**, 359–370 (2018).
17. Memristor-based circuit implementations of recognition network and recall network with forgetting stages
L. Yang, Z. Zeng, **Y. Huang**, S. Wen
IEEE Transactions on Cognitive and Developmental Systems **10**, 1133–1142 (2018).

Conference Publications

1. Hardware-algorithm co-design for hyperdimensional computing based on memristive system-on-chip
Y. Huang, A.J. Rad, Q. Xia
NeurIPS 2024 Workshop Machine Learning with New Compute Paradigms, 1–8 (2024).
2. Three-dimensional memristor-based crossbar architecture for capsule network implementation.
Y. Huang, R. Hu, Z. Zeng
2018 Eighth International Conference on Information Science and Technology, 170–175 (IEEE, 2018).

Invited Talks

1. Hardware-software co-design of memristive systems for edge intelligence
Computer Science Colloquium, City university of Hong Kong, China, 2025
2. Multi-channel testing systems for efficient prototyping of emerging memory devices and arrays
EIPBN 2024 Workshop on Emerging Memory Infrastructure, La Jolla, CA, USA, 2024.

Teaching and Advising

ECE 455/555 Embedded Systems	Fall 2026
University of Tennessee, Knoxville	
Mentor for REU students	Summer 2022/2023
University of Massachusetts Amherst	Project: FPGA programming
Mentor for Senior Design Project	Fall 2018
Missouri University of Science and Technology (Remote)	Project: ML with non-volatile memory

Professional Services

Journal Reviewer for:

- *Science Advances*
- *Nature Communications*
- *Advanced Functional Materials*
- *IEEE Transactions on Cybernetics*
- *IEEE Transactions on Systems, Man and Cybernetics: Systems*
- *IEEE Transactions on Emerging Topics in Computational Intelligence*
- *IEEE Transactions on Circuits and Systems II: Express Briefs*
- *Knowledge-Based Systems*
- *Information Sciences*
- *Neural Networks*
- *Cognitive Computation*
- *Scientific Reports*

Review Committee Member:

- *2026 IEEE International Symposium on Circuits and Systems*